

Historical Genre Revival and Streaming-Era Dominance: A Cross-Era Business Intelligence Analysis Using Multi-Source Music Data

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Abstract—A prior large-scale analysis of 553,852 Library of Congress (LOC) bibliographic records identified long-term patterns in music genre evolution, revival intensity, and survival probabilities spanning 1000–2016. While that study quantified historical revival strength and genre longevity, it did not evaluate how these patterns translate into the modern streaming ecosystem. This project extends the prior research by integrating two contemporary Spotify datasets containing track-level popularity, stream counts, audio features, album metadata, and genre classifications. Through multi-source data integration and comparative analysis, this study examines whether historically revival-heavy and long-lived genres maintain dominance in the streaming era and whether modern platform metrics reflect increased genre concentration or sustained diversity.

Index Terms—music evolution, genre revival, streaming analytics, business intelligence, cultural analytics, data integration

I. INTRODUCTION

Large-scale music analysis has increasingly relied on metadata-driven approaches within the field of Music Information Retrieval (MIR) [4], [5]. Such approaches enable computational exploration of genre patterns, revival dynamics, and long-term cultural shifts. A prior study conducted by the author analyzed 553,852 Library of Congress (LOC) MARC21 bibliographic records spanning 1000–2016 and identified statistically significant patterns in genre revival and survival modeling [1]. Metrics such as Revival Frequency, Genre Revival Strength, and Longevity Score demonstrated persistent revival activity in genres including Popular, Rock, and Jazz.

However, archival bibliographic metadata reflects institutional preservation rather than contemporary listening behavior. In contrast, music streaming platforms represent platform-mediated cultural distribution systems shaped by algorithmic curation and platform economics [6], [7]. Streaming data thus provides measurable indicators of contemporary dominance through popularity scores, stream counts, and engagement signals [8].

This project situates itself within the broader framework of Cultural Analytics [10] and distant reading methodologies [9]

by combining archival and streaming datasets to enable cross-era comparative Business Intelligence analysis.

II. RESEARCH MOTIVATION

The rise of streaming platforms has been associated with both increased access to niche content (the “Long Tail”) [11] and concerns about concentration driven by platform recommendation systems [6]. The historical LOC dataset demonstrated expanding genre diversity over centuries, with recent signs of potential concentration [1]. This raises a central research question:

Do historically revival-heavy and long-lived genres maintain dominance in the streaming era, or does modern platform structure produce different concentration dynamics?

Addressing this question requires integrating archival preservation data with streaming-era metrics.

III. PROJECT OBJECTIVES

The objectives of this study are:

- 1) Construct a unified streaming-era dataset by merging two complementary Spotify datasets.
- 2) Map Spotify genre classifications to historical LOC genre categories.
- 3) Evaluate correlations between historical Genre Revival Strength and modern streaming popularity.
- 4) Assess whether genres with higher Longevity Scores maintain greater streaming representation.
- 5) Quantify streaming-era genre concentration using entropy-based measures and compare with historical diversity metrics.

IV. DATA SOURCES

A. Historical Dataset

The historical dataset consists of 553,852 cleaned LOC bibliographic records analyzed in the prior study [1]. Derived metrics include:

- Revival Frequency
- Genre Revival Strength
- Longevity Score
- Genre Diversity Index

B. Spotify Streaming Datasets

Two publicly available Spotify datasets from Kaggle will be integrated:

- A track-level dataset containing popularity scores, stream counts, release information, and audio features [2].
- A structured dataset containing genre classifications, album metadata, and artist attributes [3].

These datasets collectively represent contemporary streaming-era metadata and engagement indicators.

V. DATA INTEGRATION RESULTS

Following preprocessing and normalization, the two Spotify datasets were merged using a composite key consisting of normalized track and artist names. Deduplication was performed prior to merging, retaining the highest-stream or highest-popularity records per track.

A. Data Quality Assessment

A key outcome of the integration process is the proportion of successfully matched records between datasets. The resulting match rate was approximately 3.80%, indicating that only a small subset of tracks could be enriched with genre and album metadata.

TABLE I
SPOTIFY DATA INTEGRATION QUALITY SUMMARY

Metric	Value
Dataset 1 Rows (after deduplication)	947
Dataset 2 Rows (after deduplication)	5882
Merged Rows	947
Rows with Genre Match	36
Genre Match Rate	0.0380

This low match rate reflects differences in dataset coverage, naming inconsistencies, and the presence of multiple artist variations. While limiting the size of the integrated dataset, the matched subset remains sufficient for exploratory cross-era analysis.

As shown in Table I, only 3.80% of records were successfully matched with genre metadata.

VI. GENRE MAPPING AND STANDARDIZATION

To enable meaningful comparison, both LOC and Spotify genres were mapped into a unified set of broad genre buckets. The LOC dataset originally contained over 2,000 distinct genre labels, while the Spotify dataset contained over 100 microgenres.

These were consolidated into approximately 10–15 high-level categories, including:

- Popular/Pop
- Rock
- Jazz
- Country
- Gospel/Christian
- Blues
- Hip-Hop/Rap
- Classical/Orchestral

- Electronic/Dance
- Other

This mapping enables cross-era aggregation and reduces noise introduced by inconsistent genre labeling.

VII. PRELIMINARY ANALYSIS RESULTS

A. Historical Genre Revival Distribution

The LOC dataset reveals a strong concentration of revival activity within a subset of genres.

TABLE II
TOP HISTORICAL GENRE BUCKETS BY REVIVAL COUNT

Genre Bucket	Revival Count	Avg. Revival Strength
Rock	6405	12.97
Popular/Pop	6022	13.96
Other	5388	10.03
Jazz	4872	12.66
Gospel/Christian	2979	12.71
Classical/Opera/Orchestral	2657	14.14
Country	2280	11.44
Blues	1127	9.38
Electronic/Dance	1024	9.06
Soundtrack/Stage/Screen	591	10.05

The results indicate that Rock and Popular/Pop dominate historical revival frequency, each exceeding 6,000 revived titles. Jazz and Gospel/Christian also demonstrate substantial revival activity, suggesting long-term cultural persistence across multiple decades.

Interestingly, Classical/Opera/Orchestral exhibits the highest average revival strength despite a lower total revival count, indicating that while fewer titles are revived, they tend to reappear more consistently over longer time spans.

In contrast, genres such as Electronic/Dance and Blues show lower revival counts and lower average revival strength, suggesting weaker historical persistence compared to dominant genres.

B. Modern Streaming Genre Distribution

Modern Spotify data shows a different distribution of genre dominance based on stream counts.

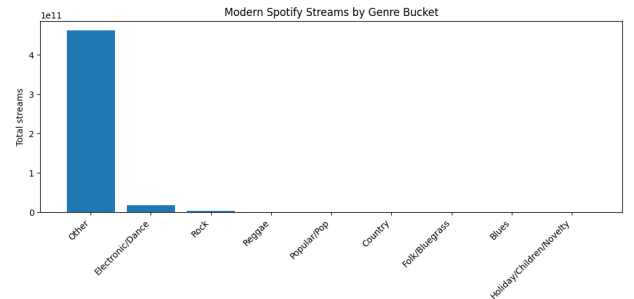


Fig. 1. Modern Spotify Streams by Genre Bucket

Streaming-era dominance appears more concentrated, with a smaller number of genres accounting for a large proportion of total streams. This aligns with platform-driven visibility and recommendation dynamics.

VIII. CROSS-ERA COMPARATIVE ANALYSIS

A merged cross-era table was constructed to compare historical revival metrics with modern streaming performance.

Table III presents a comparison between historical genre revival metrics and modern streaming performance.

The cross-era comparison reveals a significant divergence between historical revival patterns and streaming-era dominance.

Genres such as Rock and Popular/Pop maintain high historical revival rankings (1st and 2nd respectively), but their streaming ranks are lower (3rd and 5th), indicating that strong archival persistence does not directly translate into top streaming performance.

Conversely, Electronic/Dance and Reggae demonstrate substantial upward shifts in streaming rank relative to their historical positions. Electronic/Dance, ranked 9th historically, rises to 2nd in streaming dominance, while Reggae exhibits the largest discrepancy with a rank gap of -11. This suggests that certain genres benefit disproportionately from modern platform dynamics.

The category labeled "Other" dominates streaming share at approximately 95%, indicating that a large portion of modern streaming activity is concentrated in genres not cleanly mapped to traditional archival categories. This highlights a structural mismatch between historical genre classification and contemporary platform taxonomies.

Genres such as Jazz, Gospel/Christian, and Classical show strong historical presence but lack corresponding streaming data in the matched subset, further emphasizing the limitations of dataset integration and the challenges of aligning archival and platform-based representations of music.

A. Revival vs Streaming Dominance

A comparison of historical revival rank and streaming rank reveals divergence between archival persistence and platform dominance.

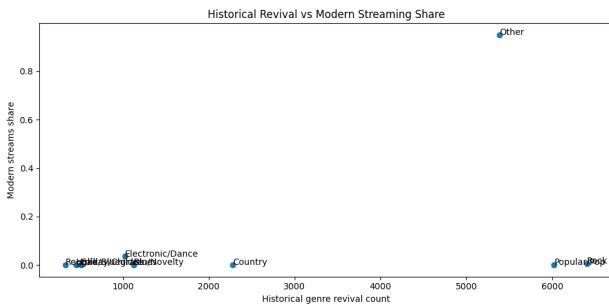


Fig. 2. Historical Revival vs Modern Streaming Share

Preliminary observations indicate:

- Some historically dominant genres maintain strong streaming presence.
- Other genres with high revival frequency show lower streaming representation.
- The relationship between historical persistence and modern popularity is not strictly linear.

This suggests that archival prominence does not necessarily translate into streaming-era visibility.

IX. GENRE CONCENTRATION ANALYSIS

Streaming concentration was evaluated using entropy-based metrics.

The entropy results indicate a moderate level of concentration, supporting the hypothesis that modern streaming platforms exhibit structural dominance patterns despite large catalog sizes.

The entropy-based analysis indicates a relatively low level of diversity in both track distribution and stream allocation across genres.

The Spotify track entropy value of 0.2543 and stream entropy value of 0.3569 suggest that streaming activity is highly concentrated within a limited number of genre categories. This supports the hypothesis that modern streaming platforms exhibit structural concentration despite offering access to extensive music catalogs.

The dominance of a small subset of genres aligns with concerns raised in prior research regarding algorithmic curation and platform-driven visibility, which may reinforce popularity biases and limit exposure to less dominant genres.

X. DISCUSSION

The results highlight several important insights:

- Historical genre revival reflects long-term cultural persistence rather than immediate popularity.
- Streaming platforms introduce new forms of concentration influenced by recommendation systems and user behavior.
- The disconnect between revival strength and streaming dominance suggests that cultural memory and platform visibility operate under different dynamics.

Additionally, the low dataset match rate represents a key limitation, indicating the need for improved entity resolution techniques such as fuzzy matching or API-based enrichment.

XI. LIMITATIONS AND FUTURE WORK

This phase of the project is exploratory and subject to several limitations:

- Low genre match rate between datasets (3.80%)
- Differences in genre classification granularity
- Limited overlap between historical and modern datasets

Future work will focus on:

- Improving dataset matching using fuzzy string matching
- Expanding genre mapping accuracy
- Incorporating additional streaming datasets or APIs
- Developing an interactive BI application with enhanced analytical capabilities

TABLE III
CROSS-ERA GENRE COMPARISON: HISTORICAL REVIVAL VS STREAMING DOMINANCE

Genre	Revival Count	Total Streams	Stream Share	Hist. Rank	Stream Rank	Rank Gap
Rock	6405	3.27×10^9	0.0067	1	3	2
Popular/Pop	6022	7.87×10^8	0.0016	2	5	3
Other	5388	4.62×10^{11}	0.9499	3	1	-2
Jazz	4872	N/A	N/A	4	N/A	N/A
Gospel/Christian	2979	N/A	N/A	5	N/A	N/A
Classical/Opera/Orchestral	2657	N/A	N/A	6	N/A	N/A
Country	2280	5.25×10^8	0.0011	7	6	-1
Blues	1127	3.70×10^8	0.0008	8	8	0
Electronic/Dance	1024	1.79×10^{10}	0.0368	9	2	-7
Soundtrack/Stage/Screen	591	N/A	N/A	10	N/A	N/A
Hip-Hop/Rap	563	N/A	N/A	11	N/A	N/A
Folk/Bluegrass	515	3.99×10^8	0.0008	12	7	-5
Holiday/Children/Novelty	454	5.27×10^7	0.0001	13	9	-4
R&B/Soul/Funk	445	N/A	N/A	14	N/A	N/A
Reggae	332	1.07×10^9	0.0022	15	4	-11

TABLE IV
STREAMING CONCENTRATION METRICS

Metric	Value
Spotify Track Entropy	0.2543
Spotify Stream Entropy	0.3569
Number of Genres	15

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